Exp. No: 12 End to End Communication at transport layer.
8/10/26.

Aim:

To implement echo client server using TCP/UDP sockets.

TCP

```
import socket
import threading
# .... Server ....
def start_server():
    server = socket.socket( Socket.AF_INET, Socket.SOCK_STREAM)
    server.bind('127.0.0.1', 5000))
    server.listen (1)
    print ("[Server] Listening on 127.0.0.1:5000")
    conn. addr = server.accept().
    print (f"[SERVER] connected by {addr}").
    while True:
        data = conn.recv (1024).decode()
        if not data or data.lower() == 'quit':
            print ("[SERVER] Connection closed by client"
            break.
        print (f"[SERVER] Received : [data]".encode())
    conn.close()
    server.close()
# .... Client ....
def start_client():
    client = socket.socket (Socket.AF_INET, socket.SOCK_STREAM)
    client.connect (('127.0.0.1', 5000)
    print ("[CLIENT] Connected to server")
```

```
    while True:
        msg = input("Enter message (or 'quit' to exit): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()

#.... MAIN....
if __name__ == "__main__":
    # Run server in a background thread.
    server_thread = threading.Thread(target = start_server,
                                     daemon = True)
    server_thread.start()
    #Run client in main thread
    start_client()
```

Output

```
====== Restart: C:/Users/tkala/Downloads/tcp.py ======
[SERVER] Listening on 127.0.0.1:5000 [CLIENT] Connected to Server
[SERVER] connected by ['127.0.0.1', 53458] Enter message (or 'quit to exit):
hello
[SERVER] Received : hello
[SERVER RESPONSE] server received: hello
Enter message (or 'quit' to exit):
[SERVER] Listening on 127.0.0.1:5000 [Client] Connected to Server
[SERVER] Connected by ('127.0.0.1', 60328) Enter message (or quit to exit):
Hi
[SERVER] Recieved .Hi
[SERVER RESPONSE) Server received: Hi
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client.
```

UDP

```
import socket
import threading

def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] client {addr} ended the
                                        connection.")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr)
    s.close()
print("[SERVER] stopped")

def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] closing connection...")
            break
        data, _ = c.recvfrom(1024)
        print("Echo from server:", data.decode())
    c.close()
print("[CLIENT] stopped.")
```

```
if __name__ == '__main__':
    threading.Thread(target=udp_server, daemon=True).start()
    udp_client()
```

Output:

```
Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.1:12000
hi
[SERVER] Received from ('127.0.0.1', 49670): hi
Echo from server: hi
Enter message (or 'quit' to stop): hello
[SERVER] Received from ('127.0.0.1', 49670): hello
Echo from server: hello
Enter message (or 'quit' to stop): how are you
[SERVER] Received from ('127.0.0.1', 49670): how are you
Echo from server: how are you
Enter message (or 'quit' to stop): quit
[Client] Closing connection... [SERVER] client ('127.0.0.1', 49670) ended the connection.
[CLIENT] Stopped. [SERVER] Stopped.
```

Result:

Implementation of TCP and UDP successfully executed.